Functional Annotation Report: *sad-I* (PP_2488, UniProt Q88K05)

NAD^+-dependent succinic semialdehyde dehydrogenase of *Pseudomonas putida* KT2440

0. Identity verification (mandatory)

Attribute	Value	Source
UniProt	Q88K05 (*Q88K05_PSEPK*, TrEMBL, PE-3 inferred from homology)	UniProt
Gene symbol	*sad-I*	UniProt / KEGG
Locus	PP_2488 (position complement 2,835,657–2,837,048)	KEGG
Protein	NAD^+-dependent succinate semialdehyde dehydrogenase, **EC 1.2.1.24**	UniProt / GenBank AAN68100
Organism	*Pseudomonas putida* KT2440 (ATCC 47054 / DSM 6125 / NCIMB 11950)	UniProt
Length / mass	463 aa / 50.3 kDa	UniProt
Family / domains	Aldehyde dehydrogenase (ALDH) superfamily; Pfam PF00171 (Aldedh); CDD cd07100 (ALDH_SSADH1_GabD1); InterPro IPR044148 (ALDH_GabD1-like), IPR047110 (GABD/Sad-like); eggNOG COG1012	UniProt / KEGG

Verification result: CONFIRMED. The gene symbol (*sad-I*), organism (*P. putida* KT2440), EC number (1.2.1.24), and ALDH-family/GabD1-like domains all match the provided target identity and are internally consistent across UniProt, KEGG and GenBank. There is no gene-symbol ambiguity: "*sad*" denotes **s**uccinic **a**ldehyde **d**ehydrogenase, and KT2440 explicitly carries two *sad* genes (*sad-I*/PP_2488 and *sad-II*/PP_3151) plus two *gabD* genes, all consistent with the SSADH function. This report describes PP_2488.

> **Evidence caveat.** No study has, to my knowledge, purified or genetically dissected PP_2488 *specifically*. The functional assignment below rests on: (i) authoritative database annotation; (ii) direct biochemistry of the *P. putida* NAD-preferring SSADH enzyme class; (iii) crystallography + mutagenesis of the close ortholog *Sad*/YneI; and (iv) my own sequence/structure bioinformatic analysis. Confidence is high for the reaction, cofactor and localization; moderate for the precise dominant physiological pathway (see §5).

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1. Summary (answer to the research question)

sad-I encodes a **soluble, cytoplasmic, NAD$^+$-preferring succinic semialdehyde dehydrogenase (SSADH; EC 1.2.1.24)** of the aldehyde-dehydrogenase superfamily. Its primary reaction is the essentially irreversible, NAD$^+$-dependent oxidation of the reactive aldehyde **succinic semialdehyde (SSA) → succinate**, feeding carbon into the TCA cycle. It is the terminal enzyme of the **GABA "shunt"** (the pathway 4-aminobutyrate → SSA → succinate) and, more generally, oxidizes SSA produced by several converging catabolic routes (GABA, polyamine/putrescine, and aromatic 4-hydroxyphenylacetate catabolism), also serving a housekeeping **aldehyde-detoxification** role. It is one of four SSADH paralogs in KT2440 and belongs to the NAD$^+$-specific *sad*/*yneI* clade, distinct from the NADP$^+$-specific *gabD* clade that sits in the dedicated GABA-nitrogen operon.

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2. Primary function: reaction catalyzed

Reaction: succinate semialdehyde + NAD$^+$ + H_2O → succinate + NADH + H$^+$ (EC 1.2.1.24; UniProt/KEGG annotation; KEGG Orthology K00135).

SSADHs are ubiquitous enzymes that "catalyze the oxidation of succinic semialdehyde (SSA) to succinic acid in the presence of NAD(P)$^+$" (Jang et al., *Biochem. Biophys. Res. Commun.* 2015; P 25888791).

Because the aldehyde is oxidized to a carboxylate with concomitant hydride transfer to NAD^+, the reaction is thermodynamically favorable and physiologically unidirectional (biosynthetic/ degradative "clean-up" direction).

Catalytic mechanism (inferred from the fold and conserved residues). The ALDH mechanism proceeds via a catalytic cysteine nucleophile that attacks the aldehyde carbonyl to form a thiohemiacetal, hydride transfer to NAD^+, and hydrolysis of the resulting thioester, with a conserved glutamate acting as general base. My alignment of sad-I to the crystallographically characterized ortholog *Sad*/YneI (see §4) locates all of these elements:

- **Catalytic nucleophile Cys265** (in the "G-Q-V-**C**-I-S-S" motif), equivalent to the catalytic Cys268 of *Salmonella*/E. coli YneI whose side chain sits next to the NAD^+ nicotinamide ring (Zheng et al., *FEBS J.* 2013; P 23229889).
- **Catalytic general base Glu362** (≡ YneI Glu365).
- **Substrate-orienting Trp133** (≡ YneI Trp136).
- ALDH cofactor/substrate-binding motifs EPWNFP (131), TGSEGAG (208) and ELGGAD (231).

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3. Substrate specificity and cofactor preference

Cofactor: NAD^+-preferring. The gene is annotated NAD^+-dependent (EC 1.2.1.24). This is supported by two independent lines of evidence: - **Biochemistry of the enzyme class in *P. putida*.** Sànchez et al. (*Biochem. J.* / Int. J. Biochem. 1988; P 3355840) separated two SSADHs from *P. putida*. The smaller dimeric enzyme (≈100 kDa; 53 kDa subunit) "acts preferentially with NAD but reduces NADP at 9% of the rate observed for NAD," i.e., a strongly NAD-preferring SSADH — the biochemical counterpart of the *sad* class to which PP_2488 belongs. (The larger ≈200 kDa enzyme was NADP-dependent and putrescine-induced — the *gabD* counterpart.) - **Conserved cofactor determinant.** The residue shown by mutagenesis to confer NAD^+ preference in YneI (**Lys160**) is conserved in sad-I as **Lys157** (P 23229889 ; my alignment).

Substrate: succinic semialdehyde, with promiscuity toward related semialdehydes. The physiological substrate is SSA (a 4-carbon ω-aldehyde-carboxylate). NAD-specific SSADHs of this class are, however, catalytically promiscuous: - Classic work on the *E. coli* NAD- vs NADP-specific SSADHs found that "the NADP-specific enzyme catalyzes only the oxidation of succinate-semialdehyde... whereas the NAD-specific form is active also towards n-butyraldehyde" (Cozzani et al. 1980; P 7004491) — i.e., the NAD-type enzyme (sad-I's class) has broader aldehyde tolerance. - In engineered *E. coli*, the native *Sad* had to be deleted because it oxidizes **malate semialdehyde**, a non-native C4 semialdehyde, diverting flux from 2,4-dihydroxybutyrate production (Nguyen et al. 2025; P 40421117). - SSADH promiscuity has been repeatedly recruited during adaptive laboratory evolution to repair distinct metabolic deficiencies (He et al., 2024; P 39406738).

This promiscuity underlies a general **detoxification** function: SSADHs "play an important role in... the detoxification of accumulated SSA" (P 25888791).

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4. Structural and evolutionary evidence

- **Fold/oligomer.** sad-I is a two-domain (Rossmann-like NAD-binding + catalytic) ALDH-superfamily protein (Gene3D 3.40.605.10 / 3.40.309.10; SSF53720). ALDH-family SSADHs function as homodimers/homotetramers; the *P. putida* NAD-preferring SSADH was characterized as a ≈100 kDa dimer of 53 kDa subunits (P 3355840).
- **Orthology.** By global alignment, sad-I is most similar to *E. coli* **Sad/YneI** (P76149, the NAD^+-specific SSADH; **41.9–42.4 % identity**) and less similar to *E. coli* **GabD** (P25526, $NADP^+$-specific; 37.5 %). *E. coli* Sad is explicitly "the succinate-semialdehyde dehydrogenase, Sad (also known as YneI)" (Rodionova et al. 2022; P 35508583).

- **Paralog family in KT2440.** Four SSADHs (all KO K00135): NAD-type **sad-I/PP_2488** and **sad-II/PP_3151** (43.6 % mutual identity) form one clade; NADP-type **gabD-I/PP_0213** and **gabD-II/PP_4422** (53.4 % mutual identity) form the other. gabD-I sits immediately downstream of **gabT/PP_0214** (4-aminobutyrate aminotransferase), i.e., in the canonical GABA-degradation operon; **sad-I does not** — it lies in a cluster of other aldehyde/xenobiotic-processing enzymes (adjacent aldehyde dehydrogenase PP_2487; OYE-family N-ethylmaleimide reductase PP_2486; xenobiotic reductases).
- **Active-site conservation** (my alignment to Sad/YneI): catalytic **Cys265** (≡Cys268), general base **Glu362** (≡Glu365), substrate **Trp133** (≡Trp136), NAD^+-preference **Lys157** (≡Lys160). All four are conserved (one YneI residue, Asp426, aligns to Arg423 near an indel — the only divergence).

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5. Biological process / pathway context

The unifying theme is that **succinic semialdehyde is a metabolic hub aldehyde** produced by several catabolic routes, and sad-I oxidizes it to succinate for entry into the TCA cycle:

1. **GABA shunt (4-aminobutyrate degradation).** GABA is transaminated (GABA aminotransferase) to SSA, which SSADH then oxidizes to succinate — the terminal, committed step (established across taxa; human SSADH, Kim et al. 2009, P 19300440; *E. coli* NAD-SSADH induced on GABA as sole N-source, P 7004491). KEGG maps PP_2488 to *butanoate metabolism* and *alanine/aspartate/glutamate metabolism* (GABA node).
2. **Polyamine (putrescine) catabolism.** Putrescine degradation converges on GABA/SSA. In *P. putida* the NADP-dependent SSADH is putrescine-induced (P 3355840); the NAD-preferring sad-I provides an alternative/parallel SSA-oxidizing activity.
3. **Aromatic 4-hydroxyphenylacetate (4-HPA) / tyrosine catabolism.** The 4-HPA *meta*-cleavage pathway yields SSA as an end product; notably, the *P. putida* NAD-preferring SSADH is specifically **induced by growth on 4-hydroxyphenylacetate**

(P 3355840),
and KEGG lists PP_2488 under *tyrosine metabolism*. This is the most distinctive candidate physiological role separating the *sad* enzymes from the GABA-operon *gabD*.

4. **General aldehyde detoxification.** Given its promiscuity and its genomic neighborhood of aldehyde/reductase genes, sad-I also acts as a housekeeping scavenger of reactive C4 (and related) semialdehydes (P 25888791 P 40421117 P 39406738).

Regulatorily, in *E. coli* the *sad*/*yneI* gene is controlled by the LysR-type regulator **PtrR/YneJ** (P 35508583), distinct from the GabR/σ-based control of *gabTD*, reinforcing that *sad*- and *gabD*-type SSADHs are separately deployed.

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6. Localization

sad-I is a **soluble cytoplasmic** enzyme. Bioinformatic analysis found no transmembrane segment (max Kyte-Doolittle hydropathy 0.96, below the ~1.6–2.0 TM threshold; zero windows >1.6), a net-hydrophilic GRAVY of −0.071, and a polar N-terminus with no signal-peptide core; UniProt reports no signal peptide, transmembrane region or targeting sequence. Its substrate SSA is generated by soluble cytoplasmic transaminases/ring-cleavage enzymes, consistent with cytosolic action. (In eukaryotes the orthologous SSADH is mitochondrial-matrix, P 19300440; bacteria lack that compartment and the enzyme is cytosolic.)

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7. Supported and refuted hypotheses

Supported - H1 — sad-I is a NAD^+-dependent SSADH catalyzing SSA→succinate. *(DB annotation + conserved catalytic Cys265/Glu362 + NAD-preference Lys157 + class biochemistry.)* - H2 — sad-I belongs to the NAD-preferring *sad/yneI* clade, distinct from NADP-type *gabD*. *(Sequence orthology 41.9 % to Sad vs 37.5 % to GabD; clade structure of the 4 paralogs.)* - H3 — sad-I is soluble/cytoplasmic. *(No TM/signal; net-hydrophilic.)* - H4 — Broader/promiscuous aldehyde specificity + detoxification role. *(n-butyraldehyde, malate semialdehyde activity of the class.)*

Refuted / disfavored - sad-I is *not* the dedicated GABA-nitrogen operon SSADH — that role is filled by gabD-I/PP_0213 (in operon with gabT/PP_0214). sad-I is not gene-linked to gabT. - sad-I is not a membrane or secreted protein.

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8. Limitations and future directions

- **No PP_2488-specific enzymology.** Km/kcat, exact cofactor ratio, oligomeric state and substrate scope of the *specific* protein are inferred, not measured. Recombinant expression + steady-state kinetics on SSA, glutarate-semialdehyde, malate-semialdehyde and n-butyraldehyde would define specificity directly.
- **sad-I vs sad-II division of labor** is unresolved; a clean knockout/complementation and growth phenotyping on GABA, putrescine and 4-HPA would assign the dominant in-vivo pathway.
- **Structure.** An AlphaFold model exists (AlphaFoldDB Q88K05); experimental structure with NAD^+/SSA would confirm the modeled active site.
- KEGG additionally lists the generic K00135 ortholog under *lysine degradation* (glutarate-semialdehyde dehydrogenase, EC 1.2.1.20); whether sad-I contributes to glutarate metabolism in vivo (vs the dedicated CoA-independent pathway) remains to be tested.

Key references

- P 3355840
— Two SSADHs in *P. putida* (NAD-preferring vs NADP; induction by 4-HPA vs putrescine).

- P 7004491
  — *E. coli* NAD- vs NADP-specific SSADH; substrate range; GABA induction.
- P 23229889
  — Crystal structure + mutagenesis of NAD(P)$^+$ SSADH YneI (catalytic Cys268, Trp136, Glu365; Lys160 NAD$^+$ preference).
- P 25888791
  — SSADH reaction and detoxification role.
- P 19300440
  — SSADH as terminal GABA-degradation enzyme (human, mitochondrial).
- P 35508583
  — *E. coli* Sad ≡ YneI; PtrR/YneJ regulation.
- P 40421117

  P 39406738
  — SSADH substrate promiscuity.
- P 12534463
  (Nelson 2002), 26913973 (Belda 2016) — KT2440 genome/annotation.